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Exercise 2:

In order to apply these modifications, we modified the code's state machine so that it could manage both sessions. To write and read even addresses, we introduced special states, and to write and read odd addresses, we added distinct states. As a result, the test's phases were all executed in the correct order, and the address lines changed during the write and read operations in the correct order. For the purpose of switching between write and read mode as necessary, we managed the write enable signal. We set the write enable signal suitably to permit writing to the SRAM during the write cycles. We turned off writing to enable reading from the SRAM during the read cycles. Utilizing the bitwise complement of the current address, we computed the anticipated data during read operations. Because the address was modified inside a single clock cycle, we had to make adjustments to the expected data in certain circumstances to accommodate for timing discrepancies. By making this change, it was made sure that the data being read from the SRAM matched the expected data exactly and avoided erroneous mismatch detections. These changes satisfied the criterion that each memory location be adequately tested by guaranteeing that each memory location was written to and read from exactly once. Because it gave each address a consistent and distinct value, using the bitwise complement of the address as the data pattern made verification simple and trustworthy. By modifying the addressing order to increase for even addresses and decrease for odd ones, we were able to adhere to the test's prescribed guidelines. To sum up, the code now writes and reads back data in the necessary patterns and orders, exhaustively testing every place of the external SRAM. By comparing the read data with the predicted values, it properly finds any disparities and, if needed, raises the BIST_mismatch flag. These modifications improve the memory testing procedure's dependability while satisfying the objectives.

